

Hydraulic System Maintenance Report

Generated: 2026-04-08 22:24:13

Executive Summary

Total cycles analysed: 2
Cycles requiring attention: 1
Healthy cycles: 1
Overall system status: ACTION REQUIRED

System Health Overview

Pump condition: Weak leakage
Accumulator condition: Slightly reduced pressure

Cycle 1 - Detailed Finding

Pump Leakage Probabilities:

No leakage		18%
Weak leakage		71%
Severe leakage		11%

Accumulator Condition Probabilities:

Optimal pressure		12%
Slightly reduced pressure		68%
Severely reduced pressure		14%
Close to failure		6%

Top Contributing Sensors:

[+] Motor Power (Average) - high impact (value: 2341.5)
[+] Volume Flow 1 (Average) - moderate impact (value: 8.3)
[+] System Pressure 1 (Average) - low impact (value: 187.4)

AI Maintenance Finding:

FINDING:

Weak internal pump leakage detected with slightly reduced accumulator pressure, indicating moderate system risk.

SENSOR EVIDENCE:

Motor power consumption is elevated above normal operating baseline, consistent with the pump working harder to compensate for internal fluid bypass. Volume flow readings are below expected values, confirming reduced pump delivery efficiency.

CONSEQUENCE OF DEFERRAL:

Continued operation without intervention will accelerate seal degradation, progressing to severe leakage and potential unplanned shutdown.

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RECOMMENDED ACTION:

Within 24 hours - inspect pump shaft seals and check accumulator pre-charge pressure. Raise maintenance work order referencing ISO 4413 section 5.4.

Regulatory Reference

- ISO 4413 - Hydraulic fluid power: General rules and safety requirements
- API 686 - Recommended Practice for Machinery Installation and Installation Design
- ISO 18435 - Diagnostics and prognostics for hydraulic systems

All maintenance actions should be documented in the facility CMMS and approved by the maintenance superintendent before execution.